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Giant chess game enhances spatial navigational skills in 6-years-old children: preliminary findings

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ABSTRACT

The game of chess is a valuable extracurricular activity for children, with positive effects on their cognitive skills and academic achievements. We investigated the extent to which the Giant Chess Game (GCG) played on a giant chessboard enhances working memory in “navigational-vista” space and “reaching” space. We also assessed if the GCG enhances mental rotation skills. For 10 weeks, 15 children (GCG group) were involved in a giant chess class, while 15 gender and age-matched children were involved in standard didactics (control group—CG). Children were tested twice, before (T0) and after (T1) the GCG, by tasks aimed at measuring: visuo-spatial working memory (VSWM) in the navigational-vista space (Walking Corsi test); VSWM in the reaching space (Corsi Block-Tapping task); mental rotation (Rotating Flowers test). We found that the GCG group significantly improved its performance more than the CG in VSWM in both navigational-vista space and reaching space, as well as in mental rotation. Our results suggest that the GCG has positive effects on visuo-spatial abilities underlying topographical skills. Therefore, the training using GCG can help enhancing spatial ability and may have a role in contrasting the spreading of navigational deficits such as the Developmental Topographical Disorientation (DTD).

KEYWORDS

Chess players; navigation training; spatial navigation; spatial representation; topographic memory; topographic skills

Introduction

Spatial abilities are a key aspect of development. Children with high spatial abilities further explore the surrounding world in order to reach their autonomy (for a review see Pallrand & Seeber, 1984; Piccardi et al., 2019; Pribyl & Bodner, 1987; Uttal & Cohen, 2012; Uttal et al., 2013). At school, high visuo-spatial skills are considered associated with high performance in several educational domains, especially with the STEM disciplines (Science, Technology, Engineering, Mathematics) (Khine, 2016). Despite this evidence, spatial skills received less attention than other skills (e.g., reading, writing, and problem solving). In particular, the training programs in topographical orientation were used with pre-schoolers only a few times. In general, they demonstrated their usefulness in fostering the acquisition of a survey representation, which reflects a higher-level of navigational competence (e.g., Boccia et al., 2017).

The chess game can exert a slight positive influence on children's cognitive abilities, such as intelligence (Sala et al., 2017). It seems to give more advantages in learning mathematics than reading (Sala & Gobet, 2016). Playing chess requires the knowledge for many chess-specific spatial structures, such as chunks or templates (Chase & Simon, 1973; Gobet & Simon, 1996a). Moreover, the chess game requires

the ability to process the chess problem-space (the chess-board) while generating and evaluating chess positions, planning and mentally imaging a sequence of moves (Chase & Simon, 1973; Gobet & Simon, 1998a). Following Frydman and Lynn (1992) “high-level chess playing requires good intelligence and strong visuo-spatial abilities” (p. 235). Indeed, chess experts have an exceptional recall of rapidly presented chessboard stimuli (Chase & Simon, 1973) and manifest a higher working memory capacity for domain-relevant stimuli (Ericsson & Kintsch, 1995; Gobet & Simon, 1996b; Gobet & Waters, 2003). Horgan and Morgan (1990) showed that their sample of elite chess players (aged around 11 years) scored higher on a visuo-spatial reasoning test (i.e., Raven's Progressive Matrices) and on an abstract reasoning test based on the ability to take different perspectives (i.e., the Piagetian Three Mountain Task). In this vein, chess necessarily involves spatial abilities, including mental rotation, because the board is rotated according to the player's color of pieces (e.g., a piece in a8 appears to the upper left or lower right depending on the color assigned) (Maass et al., 2008). Interestingly, chess cognition was found related primarily to occipital/parietal lobes, further indicating that playing chess involves visual and spatial processing, including mental rotation (see Atherton et al., 2003).

Despite these links between chess game and different skills related to spatial cognition, to the best of our knowledge no study so far considered the relationship between chess and topographic abilities, which involve a larger space dimension. Notably, it is possible to distinguish between personal or bodily space and the extra-personal space (Committeri et al., 2007). This latter can be further subdivided in two separate components: the space within the reaching distance (near extra-personal- that, in the case of chess, corresponds to the chessboard space) and the space outside the reaching distance (far extra-personal space). The former is conceived as the “grasping distance” (i.e., the space in which an individual can grasp an object), whereas the latter extends beyond our reach and considers the space within the “walking distance” (De Nigris et al., 2013; Halligan & Marshall, 1991). Moreover, within the “walking distance” it may be useful to consider the existence of a navigational-vista space. This latter corresponds to the portion of the navigational space that can be experienced from a single location or by only little exploratory movements (Wolbers & Wiener, 2014).

Thus, the relation between the chess game and spatial ability can be more complex if we consider different space dimensions. In this direction, given that chess game is related to VSWM in the reaching space (the space of the chessboard), and VSWM in the reaching space in turn plays a key role in the topographic ability (which takes place in the navigational-vista space), it can be hypothesized that playing chess can help in improving spatial abilities also in the navigational-vista space.

Indeed, many different studies demonstrated the relationship between VSWM in the reaching space and topographic abilities such as wayfinding (e.g., Fenner et al., 2000; Meneghetti et al., 2021; Piccardi et al., 2019), and homing behavior (Piccardi et al., 2019). In general, individuals that perform better in VSWM are more accurate also in navigational tasks (Meneghetti et al., 2021). Accordingly, Nori et al. (2009) found that individuals with high VSWM performed better than individuals with low VSWM in a navigation task in a large-scale environment. Fenner et al. (2000) found that young children (5–6 years old) with high visuo-spatial abilities (i.e., VSWM, mental rotation, and mental folding) showed a better wayfinding than children with low visuo-spatial abilities.

Yet, also mental rotation, another key component of chess game, was found related to topographical abilities. Nori et al. (2009) highlighted that walking through an environment requires a continuous changing of the own perspective at each turn, that is the ability to rotate the mental representation of the environment. This means that mental rotation constituted an additional cognitive load for the spatial component of VSWM, to such an extent that individuals with low VSWM were unable to mentally represent the environment. Indeed, mental rotation allows to mentally manipulate the environmental mental representation in order to find the best route to reach a goal (Bocchi et al., 2017, 2019; Palermo et al., 2008; Piccardi et al., 2017).

Therefore, considering the link between the chess game and visuo-spatial abilities in the reaching space, chess game and mental rotation, and between visuo-spatial abilities in the reaching space and topographic abilities, the present study was aimed at investigating if visuo-spatial abilities in both reaching and navigational-vista spaces can be improved in children by a training developed on a giant chessboard.

To this purpose, two different groups of 6-years old children were compared in working memory of both reaching and navigational-vista spaces, and in mental rotation, before and after the training carried out by the giant chess game (GCG). Whereas the GCG group underwent the training sessions, the control group (CG), underwent only standard didactics. The GCG class was based on the Pompa et al.'s (2017) teaching programme. This training was implemented on a walkable chessboard where children (which are “real” chess-pawns) directly learnt some typical chess moves and memorized both patterns of position of the chess pawns and the right sequence of locations to reach a square on the chessboard. Taking place in the navigational-vista space, this GCG class was hypothesized to involve either VSWM in reaching and in extra-personal space, as well as mental rotation, given that children changed their perspective at each position on the chessboard. We hypothesized that children who underwent the training could improve their spatial abilities with respect to the CG, which underwent normal didactics. Considering the relationship between chess game and VSWM in the reaching space, on the one hand, and the relationship between VSWM in the reaching space and spatial navigation (which takes place in the navigational-vista space), on the other hand, we expected to find an improvement in the VSWM in both the reaching and navigational-vista space. Moreover, given that mental rotation plays a key role in chess game and in binding visuo-spatial memory with the reaching and navigational-vista space, we expected to find an improvement also in the mental rotation ability.

Material and methods

Participants

The study was conducted using a sample of 30 typically developing Italian children (11 M and 19 F) (mean age = 6.06 years; $SD = .25$ years age range: 6–7 years), who were recruited from the first-grade class of two primary schools (“XX Settembre” and “Carducci” primary schools) located in Terni (Italy). Inclusion criteria were defined as follows: Italian mother tongue, no primary visual or hearing impairments, no neurological conditions or emotional or behavioral problems. Children reported no experience with chess game, regardless of the version. An informal interview with their teachers allowed to ensure that all children enrolled in the study had a normal general cognitive level. The study was approved by the local ethics committee in accordance with the Declaration of Helsinki. Written and informed consent was obtained from the parents and an assent from each child. Children were subdivided into two groups: 15 children were included in the control group (CG)—6 males and 9 females—and 15 children in the Giant Chess Game group



Figure 1. The Giant Chessboard used to perform the chess training.

(GCGG)—5 males and 10 females. Children from the two groups were comparable in age [$t_{(28)} = .00$; $p = 0.1$] and gender [Chi-squared = 0.14; $p = 0.70$].

Instruments and procedure

Children were individually tested in a quiet and empty room of their school. The GCGG was tested immediately before and after having performed the GCG training, that is at T0 and T1, respectively. The CG, which underwent standard didactics, was tested at T0 and T1 without attending the GCG course, but keeping the same evaluation timeline as GCGC.

GCG training procedure

The GCG consisted of 10 chess game lessons (one per week for a total of 10-weeks) by which children learnt to play on a giant, walkable chessboard. Each lesson lasted 1 h and was divided in three working units. Lessons were taught by a Chess master with 12 years of expertise. The Chess master belonged to FIDE (the acronym for “Federation Internationale Des Echecs”), which is the International Chess Federation, that is the governing body of the sport of chess, regulating all international chess competitions.

The giant chess board (2.5 m × 2.5 m—see Figure 1) was divided in 8 rows and 8 columns of boxes, named “squares.” The rows of houses of the same color coinciding with the angles were called “diagonals.”

The giant chess game program

Work unit number 1

The activity started by introducing the chessboard to children. They were given the opportunity to discover the chessboard by themselves, walking around it. Each child received simple instructions to move on the chessboard (e.g., one step forward, two steps back, three steps left, etc...). The goal was to make the children familiar with the basic concepts of the chessboard (squares, rows and steps) and to let

them directly explore the chessboard space. During this work unit, the instructor taught the concept of the “diagonal” with few simple games. For example, if the instructor clapped his hands once, children on the chessboard had to stop, if he clapped his hands twice, children had to change the color of the “square” they were on. If children were walking diagonally on the black squares, they had to move on the white squares and vice versa.

Work unit number 2

The instructor introduced the role of the “King” (one of the pieces of the chessboard). The King can take one step in every possible direction (forward, back, left, right and diagonal). To better understand the different roles on the chessboard, children played the game in pairs. The model used here is a game called “trap the King” (using the same model, children played also “trap the bishop,” “trap the queen,” “trap the rook” and so on). The aim of this game was to let two children try to “eat” each other on the chessboard, moving like two Kings, which means moving one-step at a time. During the learning process of each “piece” children had to face “problems of dominance,” like mathematical problems. One of the most famous problem followed: “put eight queens on the chessboard avoiding these queens to attack each other.” In essence, it is an impossible problem to solve but it is very useful for learning the game. For example, one child entered the chessboard as “queen,” picking a place of his choice, then a second child entered the game and chose any place except for the “squares” controlled by the other child, and so on. It is easier to solve with eight bishops because children understand very quickly the rules that bishops can move diagonally, and by consequence that they just must put themselves horizontally or vertically in relation to the other “piece.”

Work unit number 3

This unit was dedicated to the role of the “knight” and given the peculiar nature of his movements (it moves drawing an L), children were offered a different type of game, like

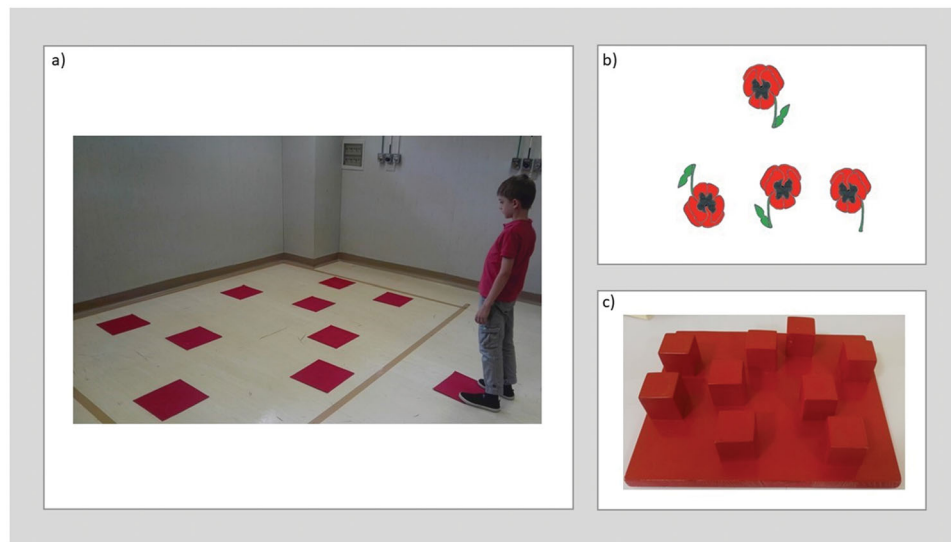


Figure 2. (a) The Walking Corsi test WalCT; (b) the Rotating Flowers test; (c) the Corsi Block Tapping Task, CBT.

“knight races” and “knight track.” “Knight races” is a very simple game whereby two children moving like the knight had to reach the other side of the chessboard. Given that the knight could not reach the end of the chessboard with exactly four steps, in order to do the last move needed without jumping out of the chessboard, children were instructed to turn around and move changing direction. Therefore, children had to understand that the knight can turn around and use its typical L movement in many ways. Instead, the “Knight Track” is the most complex part of the learning path. A child took position on a cell of the chessboard and was given a destination (another cell on the chessboard). The child, moving like a “knight” (L movement) needed to reach the exact destination of the cell.

Testing at T0 and T1

Corsi block-tapping test (CBT: Corsi, 1972; Italian version: Piccardi et al., 2013)

The CBT consists of nine wooden blocks (3×3 cm) fixed on a baseboard (30×25 cm) in a scattered array. The test allows to assess VSWM in the reaching space (Figure 2c). The experimenter tapped the series of blocks at a rate of one block every 2 s, after which the participant had to tap the same block-sequence in the same order. To obtain a visuo-spatial span, participants had to repeat the sequences of blocks of increasing length showed by the experimenter. The test started with a sequence of 2 blocks. Five trials were possible for each span number (2–9). If the participant successfully repeated 3 out of the 5 trials (consecutively or not) for a given number of length, the sequence with the next number of length was administered (ranged 2–9). For instance, if the participant successfully repeated 3 out of the 5 trials for the 2 block-sequence, the following longer trial with the 3-block sequence was administered. If the child failed in repeating 3 out of the 5 trials, the test stopped. The span was obtained by counting the number of blocks composing the last three sequences correctly repeated (max span

number = 9). Two parallel versions of the test were used, one for T0 and one for T1.

Walking Corsi Test (WalCT: Piccardi et al., 2008, 2014)

The WalCT is a larger version of the CBT ($3 \text{ m} \times 2.5 \text{ m}$; scale 1:10 of the CBT) (see Figure 2a). The WalCT assesses working memory in the navigational-vista space (route memory). Nine squares ($30 \text{ cm} \times 30 \text{ cm}$) were placed on the floor of an empty room in the same position as in the standard CBT. In addition, one additional square was placed a half meter from the line determining the perimeter of the walking area to indicate the starting position. Children were required to recall a path shown by the experimenter moving directly in the WalCT setting, beginning from the starting position and coming back at it once they reproduced the path. As a real navigational situation, the WalCT requires an active updating of both the mental representation of the environment/WalCT setting and the participant's position, because participants must continuously change their own perspective when moving from one square to another. The experimenter walked on several squares at a rate of one square per 2 s, after which the participant was asked to reproduce the square sequence by walking in the same order. The square sequences gradually increased in length (starting from a 2-square sequence to a 9-square sequence). The span was the number of squares in the longest sequence remembered correctly (square span). Five trials were possible for each square sequence. If the participant successfully performed 3 out of 5 trials (consecutively or not) for a given number of square-sequence, the examiner showed the following sequence of increased length. If the participant failed in performing 3 out of the 5 trials of the sequence, the test stopped. The walking span was obtained by counting the number of squares composing the last three paths correctly repeated (max span number = 9). Two parallel versions of the test were used, one for T0 and one for T1, respectively.

Rotating flowers (Piccardi, 2011, 2019; Pizzamiglio et al., 2017)

The Rotating Flowers test assesses children's ability of mentally rotating the picture of a flower. For each item, the experimenter showed the picture of a target flower (see Figure 2b). Children had to choose, among three alternatives, the picture that represented a rotation of the target flower. Children were allowed to cut the flower chosen and rotate it in order to check the answer. Two parallel versions of the test were used: five items were proposed either at T0 and T1.

Statistical analysis

We performed 3 one-way ANOVAs with the score of CBT, WalCT and Rotating Flowers at T0 as dependent variables and the group as independent variable to address differences in the two groups at T0. Then, in order to estimate the children's improvement due to the training beyond the individual differences that could be present during the typical development, the measurement of effectiveness of treatment was calculated for each variable (CBT span; WalCT span, Rotating Flowers) for each participant. The treatment effectiveness is a measure of improvement that adjusts for functional score at admission and reflects the proportion of potential improvement achieved during training. This measure has been successfully used in the rehabilitation practice (see Jacobson et al., 1999; Paolucci et al., 2012; Vanclay, 1991):

$$\text{effectiveness} = \frac{[\text{test score at T1} - \text{test score at T0}]}{[\text{maximum score obtained} - \text{test score at T0}] \times 100}$$

After that, three one-way ANOVAs were carried out in order to investigate the group differences in the effectiveness of treatment of CBT span, WalCT span, and Rotating Flowers score. Group was set as independent variable and the value of effectiveness of CBT span, WalCT span, and Rotating Flowers score as dependent variables. Five children, three in the GCG group and two in the CG group did not undergo the Rotating Flowers test and were excluded from the analyses of the Rotating Flowers test.

In addition, in order to investigate if the amount of improvement in the GCGG was comparable in CBT span, WalCT span and Rotating Flowers score, a repeated measure ANOVA was performed.

Results

A significant group difference at T0 was found in WalCT span ($F_{(1,28)} = 9.21$; $p = 0.005$; $\eta = 0.24$): the CG has a higher WalCT span than the GCGG. No group differences were found in CBT span ($F_{(1,28)} = .112$; $p = 0.74$; $\eta = 0.004$) and in Rotating Flowers scores ($F_{(1,28)} = 0.55$; $p = 0.46$; $\eta = 0.02$).

A significant difference in the effectiveness of treatment was observed between the GCGG and CG in WalCT span ($F_{(1,28)} = 17.94$; $p < 0.001$; $\eta = 0.46$), in CBT span ($F_{(1,28)} = 4.66$;

Table 1. Means and standard deviations (in brackets) of the raw scores of CBT (Corsi Block Tapping Task), WalCT (Walking Corsi test) Rotating Flowers for Giant Chess Game Group (GCGG) and Control group (CG) at T0 and T1.

		T0	T1
CBT span	GCGG	3.46 (0.83)	4.13 (0.35)
	CG	3.60 (1.29)	3.73 (0.79)
WalCT span	GCGG	1.06 (1.22)	2.93(0.96)
	CG	2.46 (1.30)	2.66(1.11)
Rotating Flowers score	GCGG	2.27(1.74)	4.16(0.83)
	CG	2.84(2.15)	3.38(1.38)

Table 2. Means and standard deviations of effectiveness of treatment of CBT (Corsi Block Tapping Task), WalCT (Walking Corsi test) Rotating Flowers for Giant Chess Game Group (GCG) and Control group (CG).

		N	Mean	SD
WalCT span	GCGG	15	21.95	16.58
	CG	15	2.43	6.59
CBT span	GCGG	15	10.36	14.23
	CG	15	0.76	9.68
Rotating flowers	GCGG	12	66.64	28.19
	CG	13	31.79	41.19

$p = 0.04$; $\eta = 0.42$) and in Rotating Flowers score ($F_{(1,23)} = 7.46$; $p = 0.01$; $\eta = 0.62$): the GCGG outperformed the CG in each test (raw scores of CBT, WalCT, Rotating Flowers for GCGG and CG at T0 and T1 are reported in Table 1. Means and standard deviations of effectiveness of treatment of CBT span, WalCT span, Rotating Flowers for GCGG and CG are reported in Table 2).

In addition, comparing the improvement only in the GCGG, the repeated measure ANOVA was significant ($F_{(1,14)} = 73.54$; $p < 0.001$; $\eta = 0.84$). The improvement in Rotating Flowers score was higher than the improvement in both the WalCT span ($p < 0.0001$) and CBT span ($p < 0.001$); then the improvement in WalCT span was higher than the improvement in CBT span ($p < 0.05$).

Discussion

Here, we presented the effects of the Giant Chess Game on different spatial abilities in 6-years old children. A group of children underwent the GCG training lasting 10 weeks and was compared with a control group (CG) that underwent standard didactics. Both groups were tested twice, before (T0) and after (T1) the training program or the standard didactics. Children underwent tests for VSWM in the reaching and navigational vista space, and mental rotation ability. Results showed that attending the GCG program improved all the spatial abilities tested. Interestingly, even if at T0 GCGG showed a poorer VSWM ability in the navigational-vista space than the CG, the training allowed to overcome the initial disadvantage. In addition, the Rotating Flowers performance was found to have a greater improvement than both the WalCT and the CBT performances, whereas the WalCT performance improved more than the CBT performance.

Many different studies found that the chess game and spatial abilities are related (e.g. Chase & Simon, 1973; Gobet & Simon, 1996a, 1998a; Horgan & Morgan, 1990), whereas many other studies found no significant relationship

between them (e.g., Djakow et al., 1927; Waters et al., 2002). The findings of the present study add more information to this debate (e.g., Frydman & Lynn, 1992). Indeed, although space around us can be divided in different dimensions, involving different spatial abilities (VSWM in reaching and navigational-vista space, mental rotation), chess game can potentially improve all of them. Specifically, the GCG can improve spatial abilities that underlie the spatial navigation skills: VSWM in reaching and navigational-vista space, as well as mental rotation ability. This also add new insights on the relationship between visuo-spatial and environmental information. Indeed, although visuo-spatial and environmental information are acquired and processed differently (Bianchini et al., 2014; Nemmi et al., 2013; Piccardi et al., 2008), and the competences in reaching and navigational-vista space develop at different time (earlier VSWM than topographic WM: Acredolo, 1978; Acredolo & Evans, 1980; Foreman et al., 1990; Piaget & Inhelder, 1948; Piccardi et al., 2014), these results support the view of a common mechanism that might account for both visuo-spatial and navigational abilities. On the other hand, CBT, WalCT and Rotating Flowers differently improved, so future studies should address this issue to clarify the relationship between visuo-spatial and environmental information.

Research regarding the relationships between chess and spatial abilities also suggests that these latter could play a key role in playing chess only in the early phases of learning; once the rules, positions, moves and the chessboard space are well acquired, spatial abilities would be less involved and trained (Grabner, 2014). Given that the GCGG has never undergone a chess class before, it is possible to assume that spatial abilities were particularly solicited because children had to learn for the first time the spatial relationships of the game. Probably, if spatial abilities were measured after a longer training, different results would have been found. This possibility explains why Waters et al. (2002) and Djakow et al. (1927) did not find a close relation between spatial abilities and the chess game: they assessed only expert chess players, far away from the first phases of learning. In that case it is possible that spatial abilities were, although important, no longer fundamental, being players already proficient.

These results allow also to clarify the question of the direction of the relationship between chess game and spatial abilities (see also Sala & Gobet, 2016): does playing chess improve spatial ability or vice-versa? In the present study chess game (children learnt sequences of chess pieces and moves) improved spatial abilities, that is VSWM in both reaching and navigational-vista spaces, as well as mental rotation.

Then, considering only the GCGG, we found that the improvement of VSWM in navigational-vista space was greater than the improvement in the reaching space. Notably, the GCGG showed also a higher improvement in mental rotation than VSWM in both navigational-vista and reaching space. Given that the GCG training requires to move around the space and mentally rotating the chess pawns and positions, it is not surprising that children

improved their performance at the Rotating Flowers test. Probably the GCGG improved more in mental rotation because this ability was specifically tapped by the training. However, this issue deserves more investigations.

All together these results have implications for the research regarding the educational benefits of chess game. Yet, considering that disorders involving visuo-spatial and navigational abilities (i.e., non-verbal Learning Disability; Developmental Topographical Disorientation; Developmental Coordination Disorder; Congenital aphantasia) are widespread in the healthy population, the exploration of activities that may improve such abilities becomes mandatory. For example, these results can provide new insights congenital disorders such as the aphantasia, that is a lifelong condition of reduced or absent voluntary imagery (Zeman et al., 2015), in which difficulties with mental rotation and topographical orientation have been underlined (spatial aphantasia) (Blazhenkova & Pechenkova, 2019; Palermo et al., 2022). In addition, one should not underestimate the positive effects that GCG could have on minimizing gender differences (e.g., Uttal et al., 2013) or age differences (Piccardi et al., 2020; Techentin et al., 2014) typically found in spatial abilities.

In conclusion, these preliminary findings represent an attempt to shed some light on the positive effects of GCG on visuo-spatial abilities during a critical developmental stage. The present study suggests that the implementation of specific navigation training programs based on the GCG for children and across the lifespan can be useful to promote both reaching and navigational-vista space abilities. For this reason, future studies should investigate the effects of the training based on the GCG using a larger sample, and if the same improvement in spatial domain is possible even using the standard-size chessboard. Future studies should also involve players with different degrees of expertise. It should also be clarified if spatial abilities mostly based on executive functions, such as travel planning (Bocchi et al., 2017, 2019, 2020), can also benefit from the GCG. Finally, it can be interesting to assess the amount of improvement of children at different time points of the training program.

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